.NET 8 Backend API

Entity & Database Guide

with PostgreSQL & Entity Framework Core

YT School — CMS & LMS Platform

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29+ Entities • EF Core Configurations • Migration-Ready Code

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1. Architecture Overview

The following diagram illustrates the high-level entity relationships in the system. The Topic entity serves as the central aggregate root, with content, media, AI assistant, and workflow entities radiating outward.

Entity Relationship Diagram

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          ENTITY RELATIONSHIP DIAGRAM                     │
├──────────────────────────────────────────────────────────────────────────┤
│                                                                          │
│   ┌──────────┐      1:1      ┌──────────────┐      1:1     ┌──────────────┐  │
│   │  User    │──────────────>│    Topic     │<─────────────│ VideoDetails │  │
│   └──────────┘               └──────────────┘              └──────────────┘  │
│        │                        │     │    │                ┌──────────────┐  │
│        │ 1:M              1:1 │  1:M│ 1:M│ 1:1           │ChannelDetails │  │
│        ▼                        ▼     │    │    ▼           └──────────────┘  │
│   ┌──────────┐    ┌──────────────┐    │    │  ┌────────────────┐            │
│   │TopicAppr. │    │AIAssistConf │    │    │  │ AudioOverview │            │
│   └──────────┘    └──────────────┘    │    │  └────────────────┘            │
│                      1:M │  1:M       │    │                                 │
│                      ▼   ▼            │    │                                 │
│                ┌──────┐ ┌──────┐      │    │                                 │
│                │AIExt │ │AIAgt │      │    │                                 │
│                └──────┘ └──────┘      │    │                                 │
│                                       ▼    ▼                                 │
│  ┌─────────┐ ┌─────────┐ ┌──────┐ ┌──────┐ ┌────────┐ ┌──────────┐        │
│  │  Tag    │ │Lesson   │ │QAItem│ │Quiz  │ │FlashCrd│ │TaskItem  │        │
│  └─────────┘ └─────────┘ └──────┘ └──────┘ └────────┘ └──────────┘        │
│  ┌──────────┐ ┌────────────┐ ┌──────────┐ ┌──────────┐                    │
│  │LessonOutl.│ │Infographic │ │RelatedVid│ │ResourceIt│                    │
│  └──────────┘ └────────────┘ └──────────┘ └──────────┘                    │
│                                                                            │
│  ┌──────────┐     M:M     ┌──────────────┐                                │
│  │ Curator  │<──────────>│ TopicCurator  │                                │
│  └──────────┘             └──────────────┘                                │
│                                                                            │
│  ┌──────────┐  1:M  ┌──────────────┐  1:M  ┌──────────────┐              │
│  │ Course   │──────>│ CourseModule  │──────>│ CourseChapter │──> Topic     │
│  └──────────┘       └──────────────┘       └──────────────┘              │
│                                                                            │
│  ┌──────────────┐   JSON snapshot of full Topic                            │
│  │ TopicVersion  │    for audit & rollback                                  │
│  └──────────────┘                                                          │
│                                                                            │
│  ┌──────────────┐   Self-referencing tree                                  │
│  │ MindMapNode   │──> parent_id (FK to self)                                │
│  └──────────────┘                                                          │
└──────────────────────────────────────────────────────────────────────────┘
```

Key relationship patterns:

- Topic is the central aggregate root — all content entities reference TopicId as FK
- VideoDetails and ChannelDetails have 1:1 relationship with Topic
- AIAssistantConfig has 1:1 relationship with Topic, using jsonb for complex nested objects
- TopicCurator is a M:M join table between Topic and Curator
- MindMapNode uses self-referencing FK (ParentId) for tree structure
- Course → CourseModule → CourseChapter forms a 3-level hierarchy
- TopicVersion stores full JSON snapshots for versioning/rollback

2. Enumerations

All enumerations are registered as PostgreSQL enum types using Npgsql. This ensures type safety at the database level.

```
public enum TopicStatus { Draft, InReview, Approved, Published, Retired }
public enum ClassificationStamp { Topic, Project }
public enum TopicLevel { Foundational, Advanced }
public enum ApprovalStatus { Pending, Approved, Rejected, NeedsRevision }
public enum AIPersona { Tutor, Coach, Expert, Mentor, Assistant }
public enum AITone { Professional, Friendly, Encouraging, Socratic, Concise }
public enum ScopeBoundary { Strict, Related, Flexible }
public enum ResponseLength { Brief, Moderate, Detailed, Comprehensive }
public enum ExternalResourceType { Documentation, Article, Video, Tool }
public enum ResourceCategory { AdditionalReadings, Downloads, Repositories, Playgrounds }
public enum UserRole { Admin, Editor, Reviewer, Curator }
```

PostgreSQL Enum Registration

```
// In Program.cs or DbContext.OnConfiguring
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseNpgsql(connectionString, npgsql =>
    {
        npgsql.MapEnum<TopicStatus>("topic_status");
        npgsql.MapEnum<ClassificationStamp>("classification_stamp");
        npgsql.MapEnum<TopicLevel>("topic_level");
        npgsql.MapEnum<ApprovalStatus>("approval_status");
        npgsql.MapEnum<AIPersona>("ai_persona");
        npgsql.MapEnum<AITone>("ai_tone");
        npgsql.MapEnum<ScopeBoundary>("scope_boundary");
        npgsql.MapEnum<ResponseLength>("response_length");
        npgsql.MapEnum<ExternalResourceType>("external_resource_type");
        npgsql.MapEnum<ResourceCategory>("resource_category");
        npgsql.MapEnum<UserRole>("user_role");
    }));
```

3. Core Entities

Topic

The central aggregate root entity representing a learning topic built around a video resource.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key (uuid)
`VideoId`	`string`	No	YouTube video identifier
`Category`	`string`	Yes	Primary category
`SubCategory`	`string`	Yes	Secondary category
`SubCategory2`	`string`	Yes	Tertiary category
`Title`	`string`	No	Topic display title
`Description`	`string`	Yes	Topic description (markdown)
`Overview`	`string`	Yes	Full overview content (markdown)
`ClassificationStamp`	`ClassificationStamp?`	Yes	Topic or Project
`Level`	`TopicLevel?`	Yes	Foundational or Advanced
`Rating`	`decimal?`	Yes	Average rating (0-5)
`Status`	`TopicStatus`	No	Draft, InReview, Approved, Published, Retired
`FirstLaunchDate`	`DateTime?`	Yes	Initial publication date
`LastUpdatedOnDate`	`DateTime?`	Yes	Last content update
`RetirementDate`	`DateTime?`	Yes	When topic was retired
`CreatedAt`	`DateTime`	No	Record creation timestamp
`UpdatedAt`	`DateTime`	No	Last modification timestamp
`CreatedByUserId`	`Guid`	No	FK to User who created

Relationships

- 1:1 → VideoDetails, ChannelDetails, AIAssistantConfig
- 1:M → Tags, LessonOutlines, Lessons, QAItems, QuizQuestions, FlashCards, TaskItems
- 1:M → AudioOverviewItems, Infographics, RelatedVideos, ResourceItems, ResourceFiles
- 1:M → MindMapNodes (root level), TopicApprovals, TopicVersions
- M:M → Curators (via TopicCurator join table)
- M:1 → User (CreatedByUserId)

C# Entity Class

```
public class Topic
{
    public Guid Id { get; set; }
    public string VideoId { get; set; } = string.Empty;
```

```
    public string? Category { get; set; }
    public string? SubCategory { get; set; }
    public string? SubCategory2 { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public string? Overview { get; set; }
    public ClassificationStamp? ClassificationStamp { get; set; }
    public TopicLevel? Level { get; set; }
    public decimal? Rating { get; set; }
    public TopicStatus Status { get; set; } = TopicStatus.Draft;
    public DateTime? FirstLaunchDate { get; set; }
    public DateTime? LastUpdatedOnDate { get; set; }
    public DateTime? RetirementDate { get; set; }
    public DateTime CreatedAt { get; set; }
    public DateTime UpdatedAt { get; set; }
    public Guid CreatedByUserId { get; set; }

    // Navigation properties
    public User CreatedBy { get; set; } = null!;
    public VideoDetails VideoDetails { get; set; } = null!;
    public ChannelDetails ChannelDetails { get; set; } = null!;
    public AIAssistantConfig? AIAssistantConfig { get; set; }
    public ICollection<Tag> Tags { get; set; } = new List<Tag>();
    public ICollection<LessonOutline> LessonOutlines { get; set; } = new
List<LessonOutline>();
    public ICollection<Lesson> Lessons { get; set; } = new List<Lesson>();
    public ICollection<QAItem> QAItems { get; set; } = new List<QAItem>();
    public ICollection<QuizQuestion> QuizQuestions { get; set; } = new
List<QuizQuestion>();
    public ICollection<FlashCard> FlashCards { get; set; } = new List<FlashCard>();
    public ICollection<TaskItem> TaskItems { get; set; } = new List<TaskItem>();
    public ICollection<AudioOverviewItem> AudioOverviewItems { get; set; } = new
List<AudioOverviewItem>();
    public ICollection<InfographicItem> Infographics { get; set; } = new
List<InfographicItem>();
    public ICollection<RelatedVideo> RelatedVideos { get; set; } = new
List<RelatedVideo>();
    public ICollection<ResourceItem> ResourceItems { get; set; } = new
List<ResourceItem>();
    public ICollection<ResourceFile> ResourceFiles { get; set; } = new
List<ResourceFile>();
    public ICollection<MindMapNode> MindMapNodes { get; set; } = new List<MindMapNode>();
    public ICollection<TopicCurator> TopicCurators { get; set; } = new
List<TopicCurator>();
    public ICollection<TopicApproval> TopicApprovals { get; set; } = new
List<TopicApproval>();
    public ICollection<TopicVersion> TopicVersions { get; set; } = new
List<TopicVersion>();
}
```

EF Core Configuration

```
public class TopicConfiguration : IEntityTypeConfiguration<Topic>
```

```
{
    public void Configure(EntityTypeBuilder<Topic> builder)
    {
        builder.ToTable("topics");
        builder.HasKey(t => t.Id);
        builder.Property(t => t.Id).HasDefaultValueSql("gen_random_uuid()");
        builder.Property(t => t.VideoId).IsRequired().HasMaxLength(50);
        builder.Property(t => t.Title).IsRequired().HasMaxLength(500);
        builder.Property(t => t.Description).HasColumnType("text");
        builder.Property(t => t.Overview).HasColumnType("text");
        builder.Property(t => t.Rating).HasPrecision(3, 2);
        builder.Property(t => t.Status).HasDefaultValue(TopicStatus.Draft);
        builder.Property(t => t.CreatedAt).HasDefaultValueSql("now()");
        builder.Property(t => t.UpdatedAt).HasDefaultValueSql("now()");

        builder.HasIndex(t => t.VideoId).IsUnique();
        builder.HasIndex(t => t.Status);
        builder.HasIndex(t => t.Category);

        builder.HasOne(t => t.CreatedBy)
            .WithMany(u => u.Topics)
            .HasForeignKey(t => t.CreatedByUserId);

        builder.HasOne(t => t.VideoDetails)
            .WithOne(v => v.Topic)
            .HasForeignKey<VideoDetails>(v => v.TopicId);

        builder.HasOne(t => t.ChannelDetails)
            .WithOne(c => c.Topic)
            .HasForeignKey<ChannelDetails>(c => c.TopicId);

        builder.HasOne(t => t.AIAssistantConfig)
            .WithOne(a => a.Topic)
            .HasForeignKey<AIAssistantConfig>(a => a.TopicId);
    }
}
```

VideoDetails

Stores YouTube video metadata associated with a Topic (1:1).

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic (unique)
`Title`	`string`	No	Video title from YouTube
`Source`	`string`	Yes	Video source URL

`Duration`	`string`	Yes	Duration string (e.g., PT10M30S)
`Embeddable`	`bool?`	Yes	Whether video can be embedded
`PrivacyStatus`	`string`	Yes	Public, Unlisted, Private
`PublishedOn`	`DateTime?`	Yes	Video publish date
`ViewCount`	`long?`	Yes	View count
`LikeCount`	`long?`	Yes	Like count
`AudioLanguage`	`string`	Yes	Primary audio language
`Description`	`string`	Yes	Video description
`Thumbnails`	`jsonb`	Yes	JSON object with thumbnail URLs/sizes

Relationships

- 1:1 → Topic (TopicId)

C# Entity Class

```
public class VideoDetails
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Source { get; set; }
    public string? Duration { get; set; }
    public bool? Embeddable { get; set; }
    public string? PrivacyStatus { get; set; }
    public DateTime? PublishedOn { get; set; }
    public long? ViewCount { get; set; }
    public long? LikeCount { get; set; }
    public string? AudioLanguage { get; set; }
    public string? Description { get; set; }
    public VideoThumbnails? Thumbnails { get; set; }

    public Topic Topic { get; set; } = null!;
}

// Value object stored as jsonb
public class VideoThumbnails
{
    public ThumbnailInfo? Default { get; set; }
    public ThumbnailInfo? Medium { get; set; }
    public ThumbnailInfo? High { get; set; }
    public ThumbnailInfo? Standard { get; set; }
    public ThumbnailInfo? MaxRes { get; set; }
}

public class ThumbnailInfo
{
```

```
    public string Url { get; set; } = string.Empty;
    public int Width { get; set; }
    public int Height { get; set; }
}
```

EF Core Configuration

```
public class VideoDetailsConfiguration : IEntityTypeConfiguration<VideoDetails>
{
    public void Configure(EntityTypeBuilder<VideoDetails> builder)
    {
        builder.ToTable("video_details");
        builder.HasKey(v => v.Id);
        builder.Property(v => v.Id).HasDefaultValueSql("gen_random_uuid()");
        builder.Property(v => v.Thumbnails).HasColumnType("jsonb");
        builder.HasIndex(v => v.TopicId).IsUnique();
    }
}
```

ChannelDetails

YouTube channel information associated with a Topic (1:1).

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic (unique)
`ChannelName`	`string`	Yes	Channel display name
`Handle`	`string`	Yes	YouTube handle (@name)
`Description`	`string`	Yes	Channel description
`BannerImageUrl`	`string`	Yes	Channel banner URL
`ProfilePictureUrl`	`string`	Yes	Channel profile picture URL
`JoinedOn`	`DateTime?`	Yes	Channel creation date
`SubscriberCount`	`long?`	Yes	Subscriber count
`VideoCount`	`long?`	Yes	Total video count

Relationships

- 1:1 → Topic (TopicId)

C# Entity Class

```
public class ChannelDetails
{
    public Guid Id { get; set; }
```

```
    public Guid TopicId { get; set; }
    public string? ChannelName { get; set; }
    public string? Handle { get; set; }
    public string? Description { get; set; }
    public string? BannerImageUrl { get; set; }
    public string? ProfilePictureUrl { get; set; }
    public DateTime? JoinedOn { get; set; }
    public long? SubscriberCount { get; set; }
    public long? VideoCount { get; set; }

    public Topic Topic { get; set; } = null!;
}
```

EF Core Configuration

```
public class ChannelDetailsConfiguration : IEntityTypeConfiguration<ChannelDetails>
{
    public void Configure(EntityTypeBuilder<ChannelDetails> builder)
    {
        builder.ToTable("channel_details");
        builder.HasKey(c => c.Id);
        builder.Property(c => c.Id).HasDefaultValueSql("gen_random_uuid()");
        builder.HasIndex(c => c.TopicId).IsUnique();
    }
}
```

4. Content Entities

Tag

Key-value metadata tags for topic classification and filtering.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Key`	`string`	No	Tag key (e.g., 'language', 'framework')
`Values`	`string`	No	Comma-separated values
`Description`	`string`	Yes	Tag description
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class Tag
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Key { get; set; } = string.Empty;
    public string Values { get; set; } = string.Empty;
    public string? Description { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

EF Core Configuration

```
public class TagConfiguration : IEntityTypeConfiguration<Tag>
{
    public void Configure(EntityTypeBuilder<Tag> builder)
    {
        builder.ToTable("tags");
        builder.HasKey(t => t.Id);
        builder.Property(t => t.Id).HasDefaultValueSql("gen_random_uuid()");
        builder.Property(t => t.Key).IsRequired().HasMaxLength(100);
        builder.Property(t => t.Values).IsRequired().HasMaxLength(1000);
        builder.HasOne(t => t.Topic).WithMany(tp => tp.Tags).HasForeignKey(t =>
t.TopicId);
    }
}
```

LessonOutline

Timestamped lesson outline entries for video chapters.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Timestamp`	`string`	No	Video timestamp (HH:MM:SS)
`Title`	`string`	No	Outline item title
`Description`	`string`	Yes	Outline item description
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class LessonOutline
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Timestamp { get; set; } = string.Empty;
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

Lesson

Full lesson content with timestamps, supporting rich markdown.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Timestamp`	`string`	No	Video timestamp
`Title`	`string`	No	Lesson title
`Content`	`string`	Yes	Lesson content (markdown)

`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class Lesson
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Timestamp { get; set; } = string.Empty;
    public string Title { get; set; } = string.Empty;
    public string? Content { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

QAItem

Question and answer pairs for topic comprehension.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Question`	`string`	No	The question text
`Answer`	`string`	No	The answer text (markdown)
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class QAItem
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Question { get; set; } = string.Empty;
    public string Answer { get; set; } = string.Empty;
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

QuizQuestion

Multiple-choice quiz questions with explanations.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Question`	`string`	No	Question text
`Options`	`jsonb`	No	JSON array of option strings
`CorrectAnswer`	`int`	No	Zero-based index of correct option
`Explanation`	`string`	Yes	Explanation of the correct answer
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class QuizQuestion
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Question { get; set; } = string.Empty;
    public List<string> Options { get; set; } = new();
    public int CorrectAnswer { get; set; }
    public string? Explanation { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

EF Core Configuration

```
public class QuizQuestionConfiguration : IEntityTypeConfiguration<QuizQuestion>
{
    public void Configure(EntityTypeBuilder<QuizQuestion> builder)
    {
        builder.ToTable("quiz_questions");
        builder.HasKey(q => q.Id);
        builder.Property(q => q.Options).HasColumnType("jsonb");
        builder.HasOne(q => q.Topic).WithMany(t => t.QuizQuestions).HasForeignKey(q =>
q.TopicId);
    }
}
```

FlashCard

Flash cards for spaced repetition learning.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Front`	`string`	No	Front side content
`Back`	`string`	No	Back side content
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class FlashCard
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Front { get; set; } = string.Empty;
    public string Back { get; set; } = string.Empty;
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

TaskItem

Actionable tasks/assignments for learners.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Timestamp`	`string`	Yes	Related video timestamp
`Title`	`string`	No	Task title
`Description`	`string`	Yes	Task description (markdown)
`IsCompleted`	`bool`	No	Default false
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class TaskItem
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string? Timestamp { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public bool IsCompleted { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

MindMapNode

Self-referencing tree structure for mind map visualization.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`ParentId`	`Guid?`	Yes	FK to parent MindMapNode (null = root)
`Label`	`string`	No	Node label text
`SortOrder`	`int`	No	Display order among siblings

Relationships

- M:1 → Topic
- Self-referencing: ParentId → MindMapNode.Id
- 1:M → Children (MindMapNode collection)

C# Entity Class

```
public class MindMapNode
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public Guid? ParentId { get; set; }
    public string Label { get; set; } = string.Empty;
    public int SortOrder { get; set; }

    public Topic Topic { get; set; } = null!;
    public MindMapNode? Parent { get; set; }
```

```
    public ICollection<MindMapNode> Children { get; set; } = new List<MindMapNode>();
}
```

EF Core Configuration

```
public class MindMapNodeConfiguration : IEntityTypeConfiguration<MindMapNode>
{
    public void Configure(EntityTypeBuilder<MindMapNode> builder)
    {
        builder.ToTable("mind_map_nodes");
        builder.HasKey(m => m.Id);
        builder.HasOne(m => m.Parent)
            .WithMany(m => m.Children)
            .HasForeignKey(m => m.ParentId)
            .OnDelete(DeleteBehavior.Cascade);
        builder.HasOne(m => m.Topic)
            .WithMany(t => t.MindMapNodes)
            .HasForeignKey(m => m.TopicId);
    }
}
```

5. Media Entities

AudioOverviewItem

Audio overview files in multiple languages.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`LanguageCode`	`string`	No	ISO language code (e.g., en-US)
`Language`	`string`	No	Display language name
`Title`	`string`	No	Audio title
`Url`	`string`	Yes	Audio file URL
`Length`	`string`	Yes	Duration string
`IsDefault`	`bool`	No	Whether this is the default language
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class AudioOverviewItem
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string LanguageCode { get; set; } = string.Empty;
    public string Language { get; set; } = string.Empty;
    public string Title { get; set; } = string.Empty;
    public string? Url { get; set; }
    public string? Length { get; set; }
    public bool IsDefault { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

InfographicItem

Visual infographic images associated with a topic.

Properties

Property	Type	Nullable	Description

`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Title`	`string`	No	Infographic title
`Url`	`string`	Yes	Full-size image URL
`ThumbnailUrl`	`string`	Yes	Thumbnail image URL
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class InfographicItem
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Url { get; set; }
    public string? ThumbnailUrl { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

RelatedVideo

Related video recommendations for extended learning.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Url`	`string`	No	Video URL
`Title`	`string`	No	Video title
`Description`	`string`	Yes	Video description
`ThumbnailUrl`	`string`	Yes	Video thumbnail URL
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class RelatedVideo
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Url { get; set; } = string.Empty;
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public string? ThumbnailUrl { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

ResourceItem

Categorized external resource links.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Title`	`string`	No	Resource title
`Category`	`ResourceCategory`	No	AdditionalReadings, Downloads, Repositories, Playgrounds
`Url`	`string`	No	Resource URL
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class ResourceItem
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Title { get; set; } = string.Empty;
    public ResourceCategory Category { get; set; }
    public string Url { get; set; } = string.Empty;
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

ResourceFile

Downloadable file resources (PDFs, docs, etc.).

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`Title`	`string`	No	File display title
`Url`	`string`	Yes	File download URL
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → Topic

C# Entity Class

```
public class ResourceFile
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Url { get; set; }
    public int SortOrder { get; set; }
    public Topic Topic { get; set; } = null!;
}
```

6. AI Assistant Entities

The AI Assistant configuration uses a combination of relational tables and PostgreSQL jsonb columns for complex nested objects that don't need individual querying.

AIAssistantConfig

Central AI assistant configuration for a Topic (1:1). Uses jsonb for ContentFilters and KnowledgeSources.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic (unique)
`Name`	`string`	No	Assistant display name
`Persona`	`AIPersona`	No	Tutor, Coach, Expert, Mentor, Assistant
`Tone`	`AITone`	No	Professional, Friendly, Encouraging, Socratic, Concise
`AvatarUrl`	`string`	Yes	Assistant avatar image URL
`WelcomeMessage`	`string`	Yes	Initial greeting message
`SystemInstructions`	`string`	Yes	Custom system prompt
`ScopeBoundary`	`ScopeBoundary`	No	Strict, Related, Flexible
`AllowedTopicTags`	`jsonb`	Yes	JSON array of allowed tag strings
`RestrictedTopics`	`jsonb`	Yes	JSON array of restricted topic strings
`ContentFilters`	`jsonb`	No	AIContentFilters as JSON
`KnowledgeSources`	`jsonb`	No	AIKnowledgeSources as JSON
`MaxResponseLength`	`ResponseLength`	No	Brief, Moderate, Detailed, Comprehensive
`CiteSources`	`bool`	No	Whether to cite sources in responses
`AskClarifyingQuestions`	`bool`	No	Whether AI asks clarifying questions
`ProvideExamples`	`bool`	No	Include code/concept examples
`SuggestNextSteps`	`bool`	No	Suggest learning next steps
`EncourageLearner`	`bool`	No	Encouraging tone in responses
`MaxQuestionsPerSession`	`int?`	Yes	Null = unlimited
`SessionTimeoutMinutes`	`int`	No	Session timeout in minutes
`CooldownBetweenQueries`	`int`	No	Cooldown seconds between queries
`BlockedQueryPatterns`	`jsonb`	Yes	JSON array of blocked patterns
`OutOfScopeResponse`	`string`	Yes	Response for out-of-scope questions
`UncertaintyResponse`	`string`	Yes	Response when uncertain
`EscalationEnabled`	`bool`	No	Enable human escalation

`EscalationEmail`	`string`	Yes	Escalation contact email
`SuggestRelatedTopics`	`bool`	No	Suggest related topics
`FallbackResourceUrl`	`string`	Yes	Fallback resource link
`EnableQuizMode`	`bool`	No	Enable interactive quiz mode
`EnableFlashcardMode`	`bool`	No	Enable flashcard review mode
`EnableProgressTracking`	`bool`	No	Track learner progress
`IsEnabled`	`bool`	No	Master toggle for AI assistant
`RequireAuthentication`	`bool`	No	Require user login
`AllowVoiceInput`	`bool`	No	Enable voice input
`AllowCodeExecution`	`bool`	No	Allow code sandbox execution

Relationships

- 1:1 → Topic (TopicId)
- 1:M → AIExternalResources
- 1:M → AIConnectedAgents

C# Entity Class

```
public class AIAssistantConfig
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public string Name { get; set; } = "Learning Assistant";
    public AIPersona Persona { get; set; } = AIPersona.Tutor;
    public AITone Tone { get; set; } = AITone.Friendly;
    public string? AvatarUrl { get; set; }
    public string? WelcomeMessage { get; set; }
    public string? SystemInstructions { get; set; }
    public ScopeBoundary ScopeBoundary { get; set; } = ScopeBoundary.Strict;
    public List<string>? AllowedTopicTags { get; set; }
    public List<string>? RestrictedTopics { get; set; }
    public AIContentFilters ContentFilters { get; set; } = new();
    public AIKnowledgeSources KnowledgeSources { get; set; } = new();
    public ResponseLength MaxResponseLength { get; set; } = ResponseLength.Moderate;
    public bool CiteSources { get; set; } = true;
    public bool AskClarifyingQuestions { get; set; } = true;
    public bool ProvideExamples { get; set; } = true;
    public bool SuggestNextSteps { get; set; } = true;
    public bool EncourageLearner { get; set; } = true;
    public int? MaxQuestionsPerSession { get; set; }
    public int SessionTimeoutMinutes { get; set; } = 30;
    public int CooldownBetweenQueries { get; set; }
    public List<string>? BlockedQueryPatterns { get; set; }
    public string? OutOfScopeResponse { get; set; }
    public string? UncertaintyResponse { get; set; }
    public bool EscalationEnabled { get; set; }
```

```
    public string? EscalationEmail { get; set; }
    public bool SuggestRelatedTopics { get; set; } = true;
    public string? FallbackResourceUrl { get; set; }
    public bool EnableQuizMode { get; set; }
    public bool EnableFlashcardMode { get; set; }
    public bool EnableProgressTracking { get; set; }
    public bool IsEnabled { get; set; }
    public bool RequireAuthentication { get; set; }
    public bool AllowVoiceInput { get; set; }
    public bool AllowCodeExecution { get; set; }

    public Topic Topic { get; set; } = null!;
    public ICollection<AIExternalResource> ExternalResources { get; set; } = new
List<AIExternalResource>();
    public ICollection<AIConnectedAgent> ConnectedAgents { get; set; } = new
List<AIConnectedAgent>();
}

// Value objects stored as jsonb
public class AIContentFilters
{
    public bool NoOpinions { get; set; }
    public bool NoExternalRecommendations { get; set; }
    public bool NoPersonalAdvice { get; set; }
    public bool NoPoliticalContent { get; set; } = true;
    public bool NoMedicalAdvice { get; set; } = true;
}

public class AIKnowledgeSources
{
    public bool UseOverview { get; set; } = true;
    public bool UseLessons { get; set; } = true;
    public bool UseQA { get; set; } = true;
    public bool UseQuizExplanations { get; set; } = true;
    public bool UseFlashcards { get; set; } = true;
    public bool UseResources { get; set; } = true;
    public bool UseTasks { get; set; } = true;
}
```

EF Core Configuration

```
public class AIAssistantConfigConfiguration : IEntityTypeConfiguration<AIAssistantConfig>
{
    public void Configure(EntityTypeBuilder<AIAssistantConfig> builder)
    {
        builder.ToTable("ai_assistant_configs");
        builder.HasKey(a => a.Id);
        builder.Property(a => a.Id).HasDefaultValueSql("gen_random_uuid()");
        builder.HasIndex(a => a.TopicId).IsUnique();
        builder.Property(a => a.ContentFilters).HasColumnType("jsonb");
        builder.Property(a => a.KnowledgeSources).HasColumnType("jsonb");
```

```
        builder.Property(a => a.AllowedTopicTags).HasColumnType("jsonb");
        builder.Property(a => a.RestrictedTopics).HasColumnType("jsonb");
        builder.Property(a => a.BlockedQueryPatterns).HasColumnType("jsonb");
        builder.HasOne(a => a.Topic).WithOne(t => t.AIAssistantConfig)
            .HasForeignKey<AIAssistantConfig>(a => a.TopicId);
    }
}
```

AIExternalResource

External resources the AI assistant can reference.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`AIAssistantConfigId`	`Guid`	No	FK to AIAssistantConfig
`Title`	`string`	No	Resource title
`Url`	`string`	No	Resource URL
`Type`	`ExternalResourceType`	No	Documentation, Article, Video, Tool
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → AIAssistantConfig

C# Entity Class

```
public class AIExternalResource
{
    public Guid Id { get; set; }
    public Guid AIAssistantConfigId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string Url { get; set; } = string.Empty;
    public ExternalResourceType Type { get; set; }
    public int SortOrder { get; set; }
    public AIAssistantConfig AIAssistantConfig { get; set; } = null!;
}
```

AIConnectedAgent

External AI agents that can be invoked by the assistant.

Properties

Property	Type	Nullable	Description

`Id`	`Guid`	No	Primary key
`AIAssistantConfigId`	`Guid`	No	FK to AIAssistantConfig
`Name`	`string`	No	Agent display name
`Description`	`string`	Yes	Agent description
`Endpoint`	`string`	No	Agent API endpoint URL
`IsEnabled`	`bool`	No	Whether agent is active
`SortOrder`	`int`	No	Display order

Relationships

- M:1 → AIAssistantConfig

C# Entity Class

```
public class AIConnectedAgent
{
    public Guid Id { get; set; }
    public Guid AIAssistantConfigId { get; set; }
    public string Name { get; set; } = string.Empty;
    public string? Description { get; set; }
    public string Endpoint { get; set; } = string.Empty;
    public bool IsEnabled { get; set; }
    public int SortOrder { get; set; }
    public AIAssistantConfig AIAssistantConfig { get; set; } = null!;
}
```

7. People Entities

User

System user for authentication and authorization.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`Email`	`string`	No	Unique email address
`FullName`	`string`	No	Display name
`Role`	`UserRole`	No	Admin, Editor, Reviewer, Curator
`ProfilePictureUrl`	`string`	Yes	User avatar URL
`IsActive`	`bool`	No	Account active status
`CreatedAt`	`DateTime`	No	Account creation date
`LastLoginAt`	`DateTime?`	Yes	Last login timestamp

Relationships

- 1:M → Topics (created by)
- 1:M → TopicApprovals (reviewed by)

C# Entity Class

```
public class User
{
    public Guid Id { get; set; }
    public string Email { get; set; } = string.Empty;
    public string FullName { get; set; } = string.Empty;
    public UserRole Role { get; set; } = UserRole.Editor;
    public string? ProfilePictureUrl { get; set; }
    public bool IsActive { get; set; } = true;
    public DateTime CreatedAt { get; set; }
    public DateTime? LastLoginAt { get; set; }

    public ICollection<Topic> Topics { get; set; } = new List<Topic>();
    public ICollection<TopicApproval> TopicApprovals { get; set; } = new
List<TopicApproval>();
}
```

EF Core Configuration

```
public class UserConfiguration : IEntityTypeConfiguration<User>
{
    public void Configure(EntityTypeBuilder<User> builder)
    {
        builder.ToTable("users");
```

```
        builder.HasKey(u => u.Id);
        builder.Property(u => u.Email).IsRequired().HasMaxLength(256);
        builder.HasIndex(u => u.Email).IsUnique();
    }
}
```

Curator

Subject matter expert who curates topics.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`Name`	`string`	No	Curator display name
`Handle`	`string`	Yes	Unique handle (@name)
`ProfessionalTitle`	`string`	Yes	Job title
`Description`	`string`	Yes	Full description
`DomainsOfExpertise`	`jsonb`	Yes	JSON array of expertise areas
`YearsOfExperience`	`int`	No	Years of experience
`ShortBio`	`string`	Yes	Brief biography
`JoinedOn`	`DateTime?`	Yes	When curator joined
`ProfilePictureUrl`	`string`	Yes	Profile picture URL
`Rating`	`decimal`	No	Curator rating (0-5)
`TopicCount`	`int`	No	Number of topics curated
`CourseCount`	`int`	No	Number of courses
`SkillsBadges`	`jsonb`	Yes	JSON array of skill badges
`CertificationBadges`	`jsonb`	Yes	JSON array of certifications
`EducationBadge`	`string`	Yes	Education credential
`BusinessName`	`string`	Yes	Business/company name
`WebsiteUrl`	`string`	Yes	Personal website URL

Relationships

- M:M → Topics (via TopicCurator)

C# Entity Class

```
public class Curator
{
    public Guid Id { get; set; }
    public string Name { get; set; } = string.Empty;
    public string? Handle { get; set; }
```

```
    public string? ProfessionalTitle { get; set; }
    public string? Description { get; set; }
    public List<string>? DomainsOfExpertise { get; set; }
    public int YearsOfExperience { get; set; }
    public string? ShortBio { get; set; }
    public DateTime? JoinedOn { get; set; }
    public string? ProfilePictureUrl { get; set; }
    public decimal Rating { get; set; }
    public int TopicCount { get; set; }
    public int CourseCount { get; set; }
    public List<string>? SkillsBadges { get; set; }
    public List<string>? CertificationBadges { get; set; }
    public string? EducationBadge { get; set; }
    public string? BusinessName { get; set; }
    public string? WebsiteUrl { get; set; }

    public ICollection<TopicCurator> TopicCurators { get; set; } = new
List<TopicCurator>();
}
```

EF Core Configuration

```
public class CuratorConfiguration : IEntityTypeConfiguration<Curator>
{
    public void Configure(EntityTypeBuilder<Curator> builder)
    {
        builder.ToTable("curators");
        builder.HasKey(c => c.Id);
        builder.Property(c => c.DomainsOfExpertise).HasColumnType("jsonb");
        builder.Property(c => c.SkillsBadges).HasColumnType("jsonb");
        builder.Property(c => c.CertificationBadges).HasColumnType("jsonb");
        builder.Property(c => c.Rating).HasPrecision(3, 2);
    }
}
```

TopicCurator

Many-to-many join table between Topic and Curator.

Properties

Property	Type	Nullable	Description
`TopicId`	`Guid`	No	FK to Topic (composite PK)
`CuratorId`	`Guid`	No	FK to Curator (composite PK)
`AssignedAt`	`DateTime`	No	When curator was assigned
`IsPrimary`	`bool`	No	Whether this is the primary curator

Relationships

- M:1 → Topic
- M:1 → Curator

C# Entity Class

```
public class TopicCurator
{
    public Guid TopicId { get; set; }
    public Guid CuratorId { get; set; }
    public DateTime AssignedAt { get; set; }
    public bool IsPrimary { get; set; }

    public Topic Topic { get; set; } = null!;
    public Curator Curator { get; set; } = null!;
}
```

EF Core Configuration

```
public class TopicCuratorConfiguration : IEntityTypeConfiguration<TopicCurator>
{
    public void Configure(EntityTypeBuilder<TopicCurator> builder)
    {
        builder.ToTable("topic_curators");
        builder.HasKey(tc => new { tc.TopicId, tc.CuratorId });
        builder.HasOne(tc => tc.Topic).WithMany(t => t.TopicCurators).HasForeignKey(tc =>
tc.TopicId);
        builder.HasOne(tc => tc.Curator).WithMany(c => c.TopicCurators).HasForeignKey(tc
=> tc.CuratorId);
    }
}
```

8. Course Entities

Course

A structured learning course containing modules and chapters.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`Title`	`string`	No	Course title
`Description`	`string`	Yes	Course description
`Category`	`string`	Yes	Course category
`Level`	`TopicLevel?`	Yes	Foundational or Advanced
`ImageUrl`	`string`	Yes	Course cover image URL
`Status`	`TopicStatus`	No	Draft, Published, etc.
`CreatedByUserId`	`Guid`	No	FK to User
`CreatedAt`	`DateTime`	No	Creation timestamp
`UpdatedAt`	`DateTime`	No	Last update timestamp

Relationships

- 1:M → CourseModules
- 1:M → CourseTags
- M:1 → User (CreatedByUserId)

C# Entity Class

```
public class Course
{
    public Guid Id { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public string? Category { get; set; }
    public TopicLevel? Level { get; set; }
    public string? ImageUrl { get; set; }
    public TopicStatus Status { get; set; } = TopicStatus.Draft;
    public Guid CreatedByUserId { get; set; }
    public DateTime CreatedAt { get; set; }
    public DateTime UpdatedAt { get; set; }

    public User CreatedBy { get; set; } = null!;
    public ICollection<CourseModule> Modules { get; set; } = new List<CourseModule>();
    public ICollection<CourseTag> Tags { get; set; } = new List<CourseTag>();
}
```

CourseModule

A module/section within a course.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`CourseId`	`Guid`	No	FK to Course
`Title`	`string`	No	Module title
`Description`	`string`	Yes	Module description
`SortOrder`	`int`	No	Display order within course

Relationships

- M:1 → Course
- 1:M → CourseChapters

C# Entity Class

```
public class CourseModule
{
    public Guid Id { get; set; }
    public Guid CourseId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public int SortOrder { get; set; }

    public Course Course { get; set; } = null!;
    public ICollection<CourseChapter> Chapters { get; set; } = new List<CourseChapter>();
}
```

CourseChapter

A chapter within a module, optionally linked to a Topic.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`CourseModuleId`	`Guid`	No	FK to CourseModule
`TopicId`	`Guid?`	Yes	FK to Topic (optional)
`Title`	`string`	No	Chapter title
`Description`	`string`	Yes	Chapter description
`SortOrder`	`int`	No	Display order within module

Relationships

- M:1 → CourseModule
- M:1 → Topic (optional)

C# Entity Class

```
public class CourseChapter
{
    public Guid Id { get; set; }
    public Guid CourseModuleId { get; set; }
    public Guid? TopicId { get; set; }
    public string Title { get; set; } = string.Empty;
    public string? Description { get; set; }
    public int SortOrder { get; set; }

    public CourseModule CourseModule { get; set; } = null!;
    public Topic? Topic { get; set; }
}
```

CourseTag

Tags for course classification.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`CourseId`	`Guid`	No	FK to Course
`Key`	`string`	No	Tag key
`Value`	`string`	No	Tag value

Relationships

- M:1 → Course

C# Entity Class

```
public class CourseTag
{
    public Guid Id { get; set; }
    public Guid CourseId { get; set; }
    public string Key { get; set; } = string.Empty;
    public string Value { get; set; } = string.Empty;
    public Course Course { get; set; } = null!;
}
```

9. Workflow Entities

TopicApproval

Tracks the approval workflow for topic publication.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`ReviewerUserId`	`Guid`	No	FK to User (reviewer)
`Status`	`ApprovalStatus`	No	Pending, Approved, Rejected, NeedsRevision
`Comments`	`string`	Yes	Reviewer comments
`SubmittedAt`	`DateTime`	No	When submitted for review
`ReviewedAt`	`DateTime?`	Yes	When reviewed

Relationships

- M:1 → Topic
- M:1 → User (ReviewerUserId)

C# Entity Class

```
public class TopicApproval
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public Guid ReviewerUserId { get; set; }
    public ApprovalStatus Status { get; set; } = ApprovalStatus.Pending;
    public string? Comments { get; set; }
    public DateTime SubmittedAt { get; set; }
    public DateTime? ReviewedAt { get; set; }

    public Topic Topic { get; set; } = null!;
    public User Reviewer { get; set; } = null!;
}
```

EF Core Configuration

```
public class TopicApprovalConfiguration : IEntityTypeConfiguration<TopicApproval>
{
    public void Configure(EntityTypeBuilder<TopicApproval> builder)
    {
        builder.ToTable("topic_approvals");
        builder.HasKey(a => a.Id);
        builder.HasIndex(a => new { a.TopicId, a.Status });
```

```
        builder.HasOne(a => a.Topic).WithMany(t => t.TopicApprovals).HasForeignKey(a =>
a.TopicId);
        builder.HasOne(a => a.Reviewer).WithMany(u => u.TopicApprovals).HasForeignKey(a =>
a.ReviewerUserId);
    }
}
```

TopicVersion

Stores JSON snapshots of topic state for versioning and rollback.

Properties

Property	Type	Nullable	Description
`Id`	`Guid`	No	Primary key
`TopicId`	`Guid`	No	FK to Topic
`VersionNumber`	`int`	No	Incrementing version number
`Snapshot`	`jsonb`	No	Full JSON snapshot of TopicData
`ChangeDescription`	`string`	Yes	Description of changes
`CreatedByUserId`	`Guid`	No	FK to User who created version
`CreatedAt`	`DateTime`	No	Version creation timestamp

Relationships

- M:1 → Topic
- M:1 → User

C# Entity Class

```
public class TopicVersion
{
    public Guid Id { get; set; }
    public Guid TopicId { get; set; }
    public int VersionNumber { get; set; }
    public string Snapshot { get; set; } = string.Empty;
    public string? ChangeDescription { get; set; }
    public Guid CreatedByUserId { get; set; }
    public DateTime CreatedAt { get; set; }

    public Topic Topic { get; set; } = null!;
    public User CreatedBy { get; set; } = null!;
}
```

EF Core Configuration

```
public class TopicVersionConfiguration : IEntityTypeConfiguration<TopicVersion>
{
    public void Configure(EntityTypeBuilder<TopicVersion> builder)
```

```
    {
        builder.ToTable("topic_versions");
        builder.HasKey(v => v.Id);
        builder.Property(v => v.Snapshot).HasColumnType("jsonb").IsRequired();
        builder.HasIndex(v => new { v.TopicId, v.VersionNumber }).IsUnique();
        builder.HasOne(v => v.Topic).WithMany(t => t.TopicVersions).HasForeignKey(v =>
v.TopicId);
    }
}
```

10. EF Core DbContext

The central DbContext class registers all entity sets and applies configurations from the assembly.

```
public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }

    // Core
    public DbSet<Topic> Topics => Set<Topic>();
    public DbSet<VideoDetails> VideoDetails => Set<VideoDetails>();
    public DbSet<ChannelDetails> ChannelDetails => Set<ChannelDetails>();

    // Content
    public DbSet<Tag> Tags => Set<Tag>();
    public DbSet<LessonOutline> LessonOutlines => Set<LessonOutline>();
    public DbSet<Lesson> Lessons => Set<Lesson>();
    public DbSet<QAItem> QAItems => Set<QAItem>();
    public DbSet<QuizQuestion> QuizQuestions => Set<QuizQuestion>();
    public DbSet<FlashCard> FlashCards => Set<FlashCard>();
    public DbSet<TaskItem> TaskItems => Set<TaskItem>();
    public DbSet<MindMapNode> MindMapNodes => Set<MindMapNode>();

    // Media
    public DbSet<AudioOverviewItem> AudioOverviewItems => Set<AudioOverviewItem>();
    public DbSet<InfographicItem> Infographics => Set<InfographicItem>();
    public DbSet<RelatedVideo> RelatedVideos => Set<RelatedVideo>();
    public DbSet<ResourceItem> ResourceItems => Set<ResourceItem>();
    public DbSet<ResourceFile> ResourceFiles => Set<ResourceFile>();

    // AI Assistant
    public DbSet<AIAssistantConfig> AIAssistantConfigs => Set<AIAssistantConfig>();
    public DbSet<AIExternalResource> AIExternalResources => Set<AIExternalResource>();
    public DbSet<AIConnectedAgent> AIConnectedAgents => Set<AIConnectedAgent>();

    // People
    public DbSet<User> Users => Set<User>();
    public DbSet<Curator> Curators => Set<Curator>();
    public DbSet<TopicCurator> TopicCurators => Set<TopicCurator>();

    // Course
    public DbSet<Course> Courses => Set<Course>();
    public DbSet<CourseModule> CourseModules => Set<CourseModule>();
    public DbSet<CourseChapter> CourseChapters => Set<CourseChapter>();
    public DbSet<CourseTag> CourseTags => Set<CourseTag>();

    // Workflow
    public DbSet<TopicApproval> TopicApprovals => Set<TopicApproval>();
    public DbSet<TopicVersion> TopicVersions => Set<TopicVersion>();
```

```
    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        // Apply all IEntityTypeConfiguration from assembly
        modelBuilder.ApplyConfigurationsFromAssembly(typeof(AppDbContext).Assembly);

        // Register PostgreSQL enums
        modelBuilder.HasPostgresEnum<TopicStatus>("topic_status");
        modelBuilder.HasPostgresEnum<ClassificationStamp>("classification_stamp");
        modelBuilder.HasPostgresEnum<TopicLevel>("topic_level");
        modelBuilder.HasPostgresEnum<ApprovalStatus>("approval_status");
        modelBuilder.HasPostgresEnum<AIPersona>("ai_persona");
        modelBuilder.HasPostgresEnum<AITone>("ai_tone");
        modelBuilder.HasPostgresEnum<ScopeBoundary>("scope_boundary");
        modelBuilder.HasPostgresEnum<ResponseLength>("response_length");
        modelBuilder.HasPostgresEnum<ExternalResourceType>("external_resource_type");
        modelBuilder.HasPostgresEnum<ResourceCategory>("resource_category");
        modelBuilder.HasPostgresEnum<UserRole>("user_role");

        // Global query filter for soft-delete (if implemented)
        // modelBuilder.Entity<Topic>().HasQueryFilter(t => t.Status !=
TopicStatus.Retired);
    }

    public override int SaveChanges()
    {
        UpdateTimestamps();
        return base.SaveChanges();
    }

    public override Task<int> SaveChangesAsync(CancellationToken ct = default)
    {
        UpdateTimestamps();
        return base.SaveChangesAsync(ct);
    }

    private void UpdateTimestamps()
    {
        var entries = ChangeTracker.Entries()
            .Where(e => e.State == EntityState.Modified);
        foreach (var entry in entries)
        {
            if (entry.Metadata.FindProperty("UpdatedAt") != null)
                entry.Property("UpdatedAt").CurrentValue = DateTime.UtcNow;
        }
    }
}
```

11. Initial Migration

Generate the initial migration using the EF Core CLI:

```
# Install EF Core tools
dotnet tool install --global dotnet-ef

# Add required NuGet packages
dotnet add package Npgsql.EntityFrameworkCore.PostgreSQL
dotnet add package Microsoft.EntityFrameworkCore.Design

# Create initial migration
dotnet ef migrations add InitialCreate

# Apply migration to database
dotnet ef database update
```

Key Migration Notes

- All tables use snake_case naming convention (PostgreSQL standard)
- Primary keys use gen_random_uuid() for automatic UUID generation
- jsonb columns are used for complex nested objects (thumbnails, content filters, etc.)
- PostgreSQL enums are created via HasPostgresEnum in OnModelCreating
- Composite unique indexes enforce business rules (e.g., TopicId + VersionNumber)
- Cascade delete is configured for parent-child relationships
- UpdatedAt timestamps are auto-managed via SaveChanges override

Connection String Configuration

```
// appsettings.json
{
  "ConnectionStrings": {
    "DefaultConnection":
"Host=localhost;Port=5432;Database=ytschool;Username=postgres;Password=your_password"
  }
}

// Program.cs
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseNpgsql(builder.Configuration.GetConnectionString("DefaultConnection"),
        npgsql => {
            npgsql.MapEnum<TopicStatus>("topic_status");
            // ... register all enums
        }));
```

12. Appendix: PostgreSQL-Specific Notes

UUID Generation

PostgreSQL 13+ includes gen_random_uuid() natively. For older versions, enable the uuid-ossp extension:

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
```

JSONB Indexing

For frequently queried jsonb fields, create GIN indexes:

```
CREATE INDEX ix_video_details_thumbnails ON video_details USING GIN (thumbnails);
CREATE INDEX ix_ai_config_content_filters ON ai_assistant_configs USING GIN
(content_filters);
CREATE INDEX ix_curators_domains ON curators USING GIN (domains_of_expertise);
```

Full-Text Search

Enable full-text search on topic content for the LMS search feature:

```
-- Add tsvector column
ALTER TABLE topics ADD COLUMN search_vector tsvector;

-- Create GIN index
CREATE INDEX ix_topics_search ON topics USING GIN (search_vector);

-- Update trigger to auto-populate
CREATE FUNCTION topics_search_update() RETURNS trigger AS $$
BEGIN
  NEW.search_vector :=
    setweight(to_tsvector('english', COALESCE(NEW.title, '')), 'A') ||
    setweight(to_tsvector('english', COALESCE(NEW.description, '')), 'B') ||
    setweight(to_tsvector('english', COALESCE(NEW.overview, '')), 'C');
  RETURN NEW;
END $$ LANGUAGE plpgsql;

CREATE TRIGGER topics_search_trigger
  BEFORE INSERT OR UPDATE ON topics
  FOR EACH ROW EXECUTE FUNCTION topics_search_update();
```

Performance Recommendations

- Use connection pooling (PgBouncer or Npgsql built-in pooling)
- Enable query logging in development for N+1 detection
- Use .AsNoTracking() for read-only queries
- Consider read replicas for LMS read-heavy workloads
- Implement pagination for list endpoints (keyset pagination preferred)

- Use batch operations for bulk imports (EFCore.BulkExtensions)